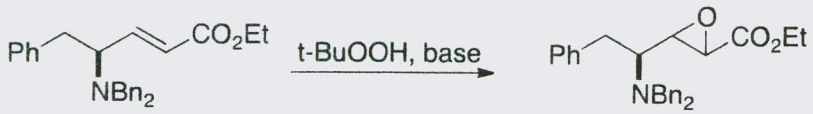
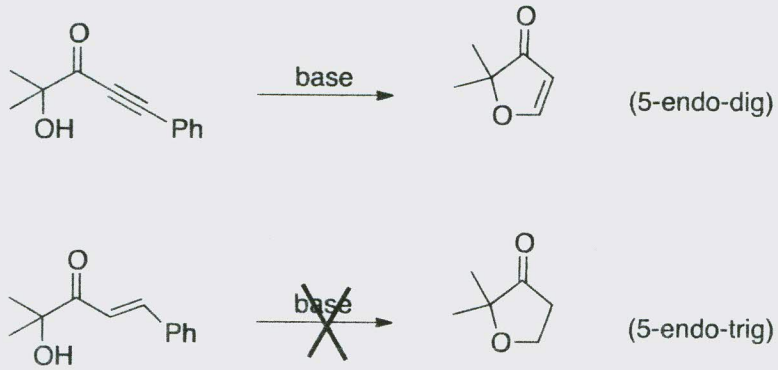
Organic Cumulative Exam
September 28, 2012

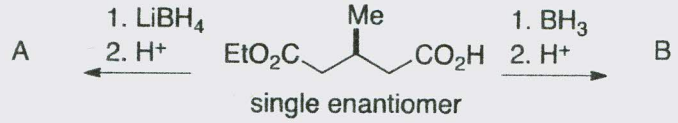
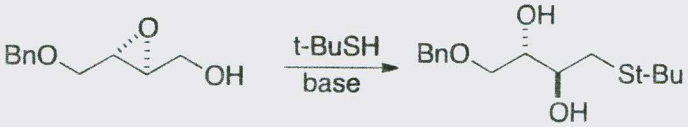
1. Use the Felkin-Anh formalism to predict the preferred diastereomer from the following reaction. Show the key intermediate involved. Consider only the trans epoxide.



2. Use orbital diagrams to rationalize the following observations:



3. What are products of the following reactions? In both cases, the H^+ in the second step is to induce ring closure to a lactone.



4. Write a mechanism for the following transformation.